Exchange rate and Production network

Macro Internal Seminar

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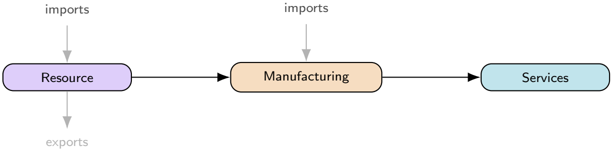
Motivation

Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index

Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**

This paper: both together — multi-sector network + FX



Questions: Optimal FX regime? Does the DC index survive a 2nd cost-push channel?

Literature

Network side (closed economy):

Rubbo (2024) — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
La'O & Tahbaz-Salehi (2022) — optimal MP in production networks
Pasten, Schoenle & Weber (2020) — sectoral rigidity heterogeneity drives MP transmission

Open-economy side:

Galí & Monacelli (2005) — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT channel)
Fanelli & Straub (2021) — FX intervention as a *second* instrument, separate from i_t
Schmitt-Grohé & Uribe (2003) — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019) — pass-through, dominant currency, IO spillovers
Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023) — frontier: managed float/fear-of-floating *optimal*, not behavioral (no network)
Gnocato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025) — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice[1]
Qiu, Wang, Xu & Zanetti (2026, JME) — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice — their own conclusion lists relaxing financial autarky as their top open extension

Gap: network + open economy + FX **regime choice** together — no one has this. Same ranking as Galí–Monacelli but a **different channel** (risk-premium, not ToT); Managed's dominance echoes Fanelli–Straub & Itskhoki–Mukhin, but from network persistence.

$\rightarrow$ paper-by-paper comparison to the literature: Appendix

[1] Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo $\rightarrow$ SOE: imported inputs Ω^F, UIP, debt-elastic premium

Ask: optimal FX regime? DC index survive with 2 cost-push channels?

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Not just Galí-Monacelli with sectors: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new

Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Model Components

Small open economy: home takes foreign price P_t^{F*}, rate i^*, and demand D_t^* as given.

1. Households — consumption/labor, foreign bond holdings
2. Firms — Cobb-Douglas technology, marginal cost
3. Production network — Leontief inverse, Domar weights
4. Firms: price setting — Calvo pricing, Phillips curve
5. Foreign sector — law of one price, UIP, NFA
6. Monetary policy — float / peg / managed float
7. Equilibrium system — the four blocks actually simulated

$\rightarrow$ full equations & notation: Appendix

Households

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize CRRA-consumption/disutility-of-labor utility, subject to a budget constraint with a domestic bond B_t (nets to zero in equilibrium) and a foreign bond B_t^*:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition; CRS makes marginal cost common across firms:

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H}\, M_{ift}^{\Omega_i^F}$$

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H}\, (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) *and* S_t (import cost); with **Calvo pricing**, only a fraction δ_i resets each period — mechanics & Phillips curve: next slide.

Production Network

Stacking each firm's input shares Ω^H_{ij}, Ω^F_i across sectors gives the **input-output network** $\Omega^H \in \mathbb{R}^{N\times N}$ and import-exposure vector $\boldsymbol{\omega}_F \in \mathbb{R}^N$.

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

applied to consumption shares $\boldsymbol{\beta}^H$ (Households) it gives each sector's **Domar weight** (total GDP sales share); applied to $\boldsymbol{\omega}_F$, its total foreign exposure:

$$\boldsymbol{\lambda}^\top = (\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}, \qquad \mathcal{M} = (I - \Omega^H)^{-1}\boldsymbol{\omega}_F$$

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H\Delta)^{-1}$, applied to labor share $\boldsymbol{\alpha}$ vs. import share $\boldsymbol{\omega}_F$:

$$\mathcal{B} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa}$$

Only relative shocks matter: $\mathcal{V}\mathbf{1} = \mathbf{0}$ — uniform TFP shocks create no cost-push; only sector-specific gaps $\boldsymbol{\chi}_t = (I - \Omega^H)^{-1}\log\mathbf{A}_t - \boldsymbol{p}_{t-1}$ do.

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t: exch. rate chg.	κ: GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Foreign Sector

P_t^{F*}, i^*, D_t^* are exogenous AR(1) processes. Law of one price:

$$P_t^{FH} = S_t P_t^{F*}$$

UIP, with a debt-elastic risk premium closing the model (Schmitt-Grohé–Uribe) and RP_t the risk-premium shock studied below:

$$i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance; exports respond to relative prices:

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \qquad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \ \rightarrow \ MC_{it} \ \rightarrow \ \pi_{it} \quad \text{(cost-push)} \qquad \hat{s}_t \ \rightarrow \ EX_t \ \rightarrow \ \tilde{y}_t \ \rightarrow \ \pi_{it} \quad \text{(demand)}$$

Monetary Policy Regimes

Three exchange-rate regimes are compared. All target the **DC index π_t^{DC}, not CPI inflation**: Rubbo (2024) shows CPI targeting does *not* achieve divine coincidence even in the closed economy — π_t^{DC} is the closed-economy-optimal benchmark, so it is the natural rule to test for survival once the economy is opened.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Everything above collapses to four blocks — this is what is actually coded up and simulated in Dynare.

Phillips curve (vector, network-amplified, two cost-push channels):

$$\boldsymbol{\pi}_t = \rho(I-\mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma+\varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

IS curve (aggregate demand; open economy adds the same relative-price term as the cost-push channel above):

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t\pi_t^C - r_t^{\mathrm{nat}}\right) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta\left(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H\right)$$

Foreign sector (UIP closes the FX block; NFA is the only state carried across regimes):

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t, \qquad \hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Policy rule (regime-dependent; the only block that changes across Float/Peg/Managed — same rules as previous slide):

$$\hat{i}_t = \begin{cases} \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t + \phi_s\hat{s}_t & \text{Managed} \end{cases}$$

where:

r_t^{nat}: natural real rate	$\hat{p}_t^H$: log home price idx.	ω, η: home bias, elast.
$\psi(\cdot), RP_t$: debt wedge, risk-prem.	π_t^{DC}: DC index (target)	$\mathcal{B}, \Gamma, \mathcal{V}$: PC coefficients

Welfare Loss Function

$$\mathcal{W} = \frac{1}{2}\Big[\underbrace{(\gamma+\varphi)\,\mathrm{Var}(\tilde{y}_t)}_{\text{output gap}} + \underbrace{\textstyle\sum_i \kappa_i\,\mathrm{Var}(\pi_{it})}_{\text{price dispersion}}\Big], \quad \kappa_i = \lambda_{D,i}\,\varepsilon\,\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Calibration: Chile Data

Parameter	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604	Data: BCCh IO table
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744	
Export share$_{1,2,3}$	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); Nakamura–Steinsson (2008)
β	0.99	Literature
γ	1.00	
φ	2.00	
η	1.50	
ε	8.00	
θ^*	2.00	
$\beta^{F,tot}$	0.10	
ψ	0.020	
ϕ_π	1.50	
ϕ_y	0.50	
ϕ_s	0.30	

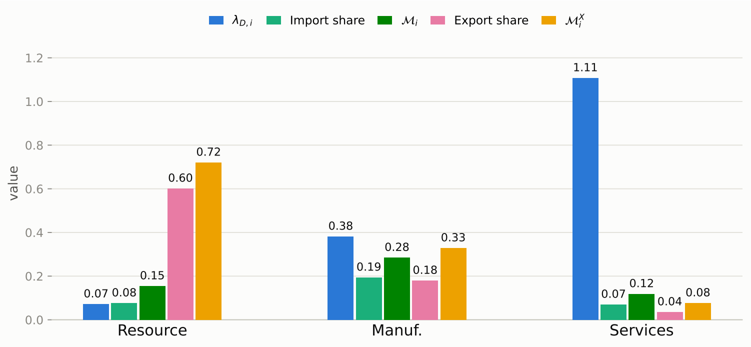
Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Calibration: Network Properties

$$\lambda_{D,i} = \left[(\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}\right]_i, \quad \mathcal{M}_i = \left[(I - \Omega^H)^{-1}\boldsymbol{\omega}_F\right]_i, \quad \mathcal{M}_i^X = \left[(I - \Omega^H)^{-1}\boldsymbol{\omega}_X\right]_i$$

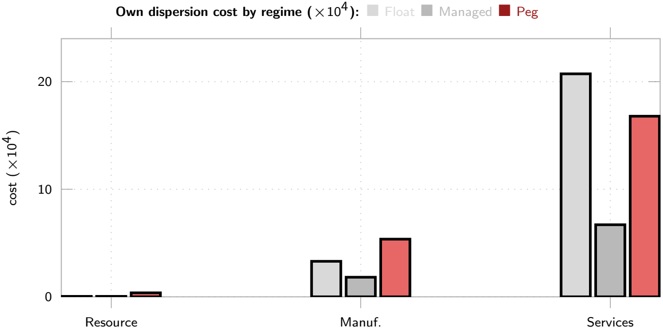
$\lambda_{D,i}$: Domar weight $\boldsymbol{\omega}_F, \boldsymbol{\omega}_X$: raw import/export share (own output) $\mathcal{M}_i, \mathcal{M}_i^X$: import/export centrality (same formula, network-propagated)



Both centralities amplify the raw share, unevenly: import exposure roughly doubles for every sector; export exposure jumps most for Manuf. (0.18→0.33), mostly via buying from Resource (60% exported). Structural fact about Ω^H; not yet fed into the simulations.

Network Exposure & Sector Welfare Preferences

Welfare weight $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ applied to each sector's own price-dispersion variance — the formal, per-sector slice of the welfare loss.

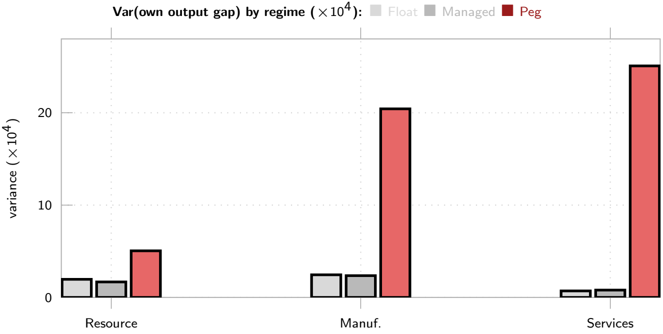


Managed dominates every sector. Peg costliest for Resource (0.37) & Manuf. (5.36); for Services, **Float is costliest** (20.73 > Peg's 16.79) — near-unit-root S_t under Float feeds the stickiest sector's marginal cost for a long time

Cost concentrates in Services under literature-sourced Calvo stickiness ($\hat{\delta}_3 = 0.03$) — Services carries most of the economy's welfare weight, so whichever regime is worst for its inflation dominates the aggregate ranking

Network Exposure: Output-Gap Variance by Sector

Sector's *own* output-gap variance — descriptive, not a welfare term (the loss function weights only the *aggregate* gap $\tilde{y}_t$).



Peg is the outlier — Float/Managed stay small and close across all sectors; Peg's variance concentrates in Manuf. and Services (20.4, 25.1), not Resource (5.0)

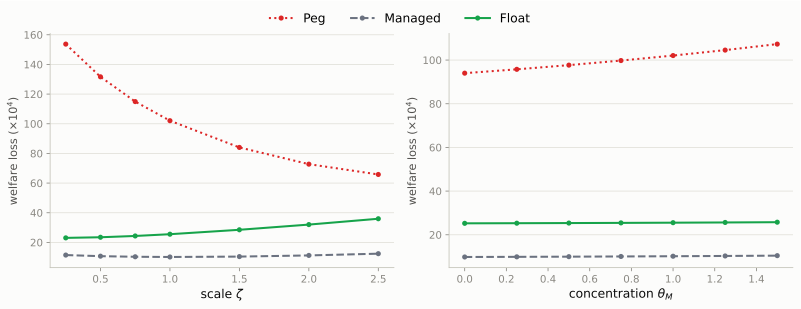
Opposite pattern from the price-dispersion chart: Peg's cost is almost entirely a *quantity* story, Float's Services cost almost entirely a *price*-dispersion one

Openness & Import Concentration

Reparametrize $\boldsymbol{\omega}_F$ only, on the REAL Chile calibration ($\alpha_i = 1 - \Omega_i^H - \omega_{F,i}$ absorbs it, holding Ω^H fixed at its Chile values):

$$\omega_{F,i}(\zeta) = \zeta\, \omega_{F,i}^{\text{base}} \qquad \omega_{F,i}(\theta_M) = \bar{\omega}_F + \theta_M \big(\omega_{F,i}^{\text{base}} - \bar{\omega}_F\big)$$

$\zeta = \theta_M = 1$: baseline. ζ: uniform rescaling. θ_M: mean-preserving redistribution across sectors (θ_M=0: uniform exposure).



Managed $<$ Float $<$ Peg robust throughout both sweeps (real Chile calibration, $\zeta = \theta_M = 1$ recovers the headline 25.47/10.17/102.05 exactly); gap still moves in **opposite directions**, though not symmetrically:

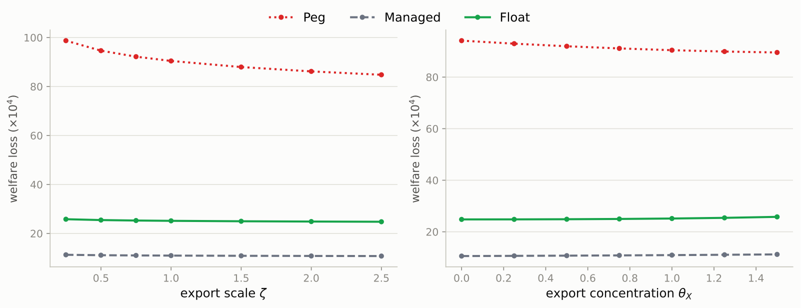
- ζ (openness): Peg $\downarrow$ (153.7 $\to$ 65.8), Float $\uparrow$ (23.1 $\to$ 36.0), Managed U-shaped (min. near ζ=1) — **converge**
- θ_M (concentration): Peg $\uparrow$ much faster (94.0 $\to$ 107.3, +14%) than Float (25.2 $\to$ 25.8, +2%) or Managed (9.8 $\to$ 10.4) — **gap still diverges**, but via Peg rising, not Float falling

Export Openness & Export Concentration

Same construction as "Openness & Import Concentration," applied to the FULL sector-export model's $\kappa_{EX,i}$ (real Chile calibration, not stylized):

$$\kappa_{EX,i}(\zeta) = \zeta\, \kappa_{EX,i}^{\text{base}} \qquad \kappa_{EX,i}(\theta_X) = \bar{\kappa}_{EX} + \theta_X \left(\kappa_{EX,i}^{\text{base}} - \bar{\kappa}_{EX}\right)$$

$\zeta = \theta_X = 1$: baseline (real export shares). ζ: uniform export-openness scaling. θ_X: mean-preserving redistribution across sectors (θ_X=0: uniform export exposure).



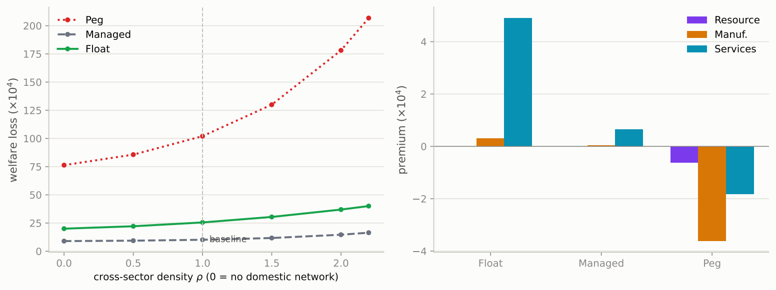
Managed < Float < Peg robust throughout both sweeps. Unlike import ζ (which moves Peg/Float in *opposite* directions — $\omega_{F,i}$ is a cost share, trading off against α_i), export ζ moves **all three regimes together**:

ζ (openness): **all regimes improve** as trade opens further — Peg most (98.7 → 84.8, −14%), Float/Managed least (≈ −4%) — since $\kappa_{EX,i}$ only enters export *demand*, not the cost side, so it never trades off against the labor share the way $\omega_{F,i}$ does

θ_X (concentration): **opposite directions, as with import concentration** — Peg ↓

Isolating the Network Channel

Real Chile calibration: scale all six cross-sector Ω^H links by ρ, holding within-sector input use & Ω^F fixed at Chile's data. ρ=0: no cross-sector domestic network. ρ=1: exact Chile baseline. Feasible to $\rho \approx 2.45$ (row 1/2 hit α_i=0) — denser than the stylized chain used previously, so the grid stops at 2.2, not 3.



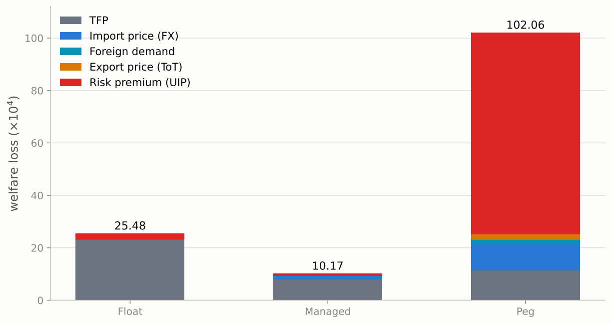
Right panel: $\kappa_i \mathrm{Var}(\pi_{it})\big|_{\rho=1} - \kappa_i \mathrm{Var}(\pi_{it})\big|_{\rho=0}$ per sector — price-dispersion only, doesn't sum to the left panel's total (sector-less output-gap term's own change not shown).

Managed is least sensitive, not neutral: ρ=0 $\rightarrow$ 1 adds $+1.17$ to Managed vs. $+5.52$ Float, $+25.72$ Peg ($20.0/9.0/76.3 \rightarrow 25.5/10.2/102.1$)

Peg's rise is a pure output-gap story: its per-sector price dispersion *falls* slightly at every sector $(-0.01/-3.62/-1.82)$ as ρ:0 $\rightarrow$ 1 — the $+25.72$ comes entirely from the output-gap term amplifying with network density, not price dispersion

Float/Managed's rise concentrates in Services $(+4.90/+0.64)$, matching Services' large

Shock Decomposition of Welfare Cost

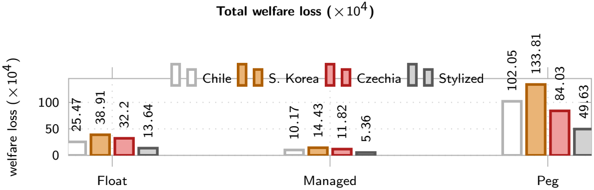


Risk-premium/UIP dominates Peg (75% of its total)

Ranking: Managed $<$ Float $\ll$ Peg — Peg no longer competitive

Result: Peg's loss is ≈78% output gap (79.53 of 102.05, zero monetary autonomy); Float's and Managed's are almost entirely price dispersion — full decomposition in appendix

Robustness: South Korea, Czechia & the Stylized Baseline



South Korea: diversified manufacturing exporter (Bank of Korea, 2022), deeper manufacturing import intensity than Chile ($\Omega^F_{\text{manuf}} = 0.24$ vs. 0.19)
Czechia: third real IO calibration (Eurostat, 2022), deepest manufacturing import intensity ($\Omega^F_{\text{manuf}} = 0.32$, EU/German supply chains)
Stylized: original hand-picked triangular Ω^H (Rubbo-style, no data) — checks the result isn't an artifact of network shape
Ranking survives in all four: Managed $<$ Float $\ll$ Peg

UIP dominates Peg's loss in all four: 75% Chile, 72% Korea, 65% Czechia, 81% stylized — not specific to one calibration. All solved nonlinear (Dynare), not linearized.

Conclusion

Managed float optimal: 10.17 vs. Float 25.47, Peg 102.05 ($\times 10^4$, Chile calibration)

Driver: risk-premium/UIP dominates Peg (75%); TFP persistence dominates Float & Managed

Optimal $\phi_s \approx 0.20$ (real Chile calibration): interior optimum, but nearly flat vs. the calibrated 0.30 ($\approx$0.3% apart) — over-stabilizing S_t beyond ≈ 0.3 is what's costly

Ranking robust to ζ, θ_M, ρ (all three now Chile-calibrated) & to Korea/Czechia calibrations; the *gap* is not

Network channel isolated (Chile calibration, ρ dial on all six cross-sector Ω^H links): ρ=0 $\rightarrow$ 1 adds +1.17 to Managed vs. +5.52 Float, +25.72 Peg — **Peg's rise is entirely an output-gap story** (its price dispersion falls slightly at every sector); Peg does *not* converge toward Float at high density, unlike an earlier stylized-network estimate

Vs. literature: same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)

2nd-order-verified (Dynare order=2, pruned, simulated): loss rises $\leq 2\%$ net of MC noise, ranking unchanged — see appendix

Next: many-sector OECD TiVA calibration (beyond the 3-sector Chile/Korea/Czechia data)

Appendix: Chile Calibration — Sector Concordance & Sources

Resource = mining/copper (Chile's export engine, 60% of own output exported) $\rightarrow$ Manuf. (mid-chain) $\rightarrow$ Services (largest value added, sticky)

$\alpha_i, \Omega^H_{ij}, \Omega^F_i, \beta^H_i$: **data** (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. 2023)

$\delta_1, \delta_2, \delta_3$: **literature** — euro-area price-change frequencies (Dhyne et al. 2005/ECB IPN) $\rightarrow$ quarterly reset prob.; cross-checked vs. Nakamura & Steinsson (2008) US PPI

$\beta, \gamma, \varphi, \eta, \varepsilon, \theta^*, \psi, \phi_\pi, \phi_y, \phi_s$: literature-standard

Sector concordance (Banco Central de Chile, CdeR 2018, 12 activities $\rightarrow$ our 3): **Resource** = {1 Agriculture, forestry & fishing, 2 Mining}; **Manufacturing** = {3 Manufacturing industry, 4 Electricity, gas, water & waste management, 5 Construction}; **Services** = {6 Trade, hotels & restaurants, 7 Transport, communications & information services, 8 Financial intermediation, 9 Real estate & housing services, 10 Business services, 11 Personal services, 12 Public administration}.

Import & export data exist at the sector level from the same tables: imports = imported intermediate input use per buying sector (sheet 21, underlies Ω^F_i); exports = exports of each sector's own output (sheet 20, underlies the export-share figures on the Calibration slide).

Appendix: Is the Ranking Robust to a Genuine 2nd-Order Solution?

Why this was a caveat: Rubbo's LQ welfare formula is 2nd-order in utility, but evaluated on *1st-order* (certainty-equivalent) paths — valid only if $\mathbb{E}[\tilde{y}_t] = \mathbb{E}[\pi_{it}] = 0$ exactly. Two open-economy features can break that: the **debt-elastic UIP wedge** (nonlinear in B_t^*) and the **near-unit-root NFA** process — both mean the true *risk-adjusted* steady state can sit away from the deterministic one.

Method: the model is already coded nonlinear (levels) in Dynare. Solve to **order 2 with pruning**, **simulate** 260k periods (analytic order-$\geq$ 2 moments are unavailable: the Taylor rule leaves price levels/S with a unit root). Compute $\mathbb{E}[X^2]$ directly (not Var) so any risk-adjusted mean shift is captured. Control for simulation noise with an identical-seed order=1 run.

Welfare loss ($\times 10^4$)	**Float**	**Managed**	**Peg**
1st-order (analytic, headline)	25.47	10.17	102.05
1st-order (simulated, same seed)	25.58	10.20	102.39
2nd-order (pruned, simulated)	**26.09**	**10.26**	**103.13**
Net 2nd-order effect	+2.0%	+0.6%	+0.7%

Ranking survives: Managed $<$ Float $<$ Peg at 2nd order, correction $\leq$ 2% everywhere

Appendix: Is the 75% Risk-Premium Share a ψ Knife-Edge?

Why this was a caveat: the "75% risk-premium" result rests on a debt-elastic UIP closing device (Schmitt-Grohé–Uribe 2003) at $\psi = 0.020$. Known pitfall: ψ near zero can create a near-unit-root NFA blowup that's a linearization artifact — the order-2 check (previous slide) ruled out spurious *nonlinearity*, not whether *this* ψ value does unreasonable work.

Method: scale ψ over 0.25×–4× baseline (0.005–0.080), re-solve all three regimes (nonlinear, Dynare), recompute the risk-premium shock's % share of welfare loss via the exact variance decomposition.

ψ	0.005	0.010	**0.020 (base)**	0.040	0.080
Peg total loss ($\times 10^4$)	138.17	120.74	**102.05**	83.69	67.46
Risk-prem. % of Peg's loss	83.0%	80.0%	**75.3%**	68.1%	57.6%

Not a knife-edge: share declines **smoothly, monotonically** across a 16× range — no blow-up near $\psi \to 0$ or the calibrated value

Mechanism robust, not the number: even at 4× baseline, risk-premium/UIP still drives a **majority** (57.6%) of Peg's loss

Ranking survives the whole grid: Managed < Float < Peg at every ψ tested

Appendix: Network vs. No-Network, on the Real Chile Data

Why this is needed: "Isolating the Network Channel" (main deck) uses a **stylized triangular** Ω^H scaled by density dial ρ, not Chile's actual **dense** Ω^H. Here: zero every Ω^H_{ij} (incl. diagonal), let α_i absorb the freed share ($\alpha_i = 1 - \omega_{F,i}$); re-derive the steady state and re-solve all three regimes on the real data.

Total welfare loss ($\times 10^4$)	**Float**	**Managed**	**Peg**
With network (baseline, headline)	25.47	10.17	102.05
No network (counterfactual)	8.13	5.30	37.94
Network premium	**+213%**	**+92%**	**+169%**

Network roughly doubles-to-triples every regime's loss — not a decorative feature, most acutely Float (+213%)

Ranking preserved either way: Managed < Float < Peg with *and* without the network

Network reallocates Peg's loss: without it, output-gap collapses (79.53 $\rightarrow$ 10.53) but price dispersion *rises* (22.51 $\rightarrow$ 27.41)

Appendix: Does Export Concentration Matter, Not Just Export Openness?

Why this is needed: the headline model allocates aggregate exports EX_t via household shares β_i^H (Services-heavy: 0.744) — *opposite* Chile's real export pattern (Resource-heavy: 0.602/0.180/0.036). FULL version: three sector-specific equations $EX_{i,t} = \kappa_{EX,i} D_t^* P_t^X (P_{i,t}/P_t^{FH})^{-\theta^*}$, $\kappa_{EX,i}$ calibrated (Newton's method) so exports hit the true exporting sector.

Total welfare loss ($\times 10^4$)	Float	Managed	Peg
Old (β^H-alloc., Services-heavy)	25.47	10.17	102.05
New (sector-specific, real data)	25.15	10.96	90.45

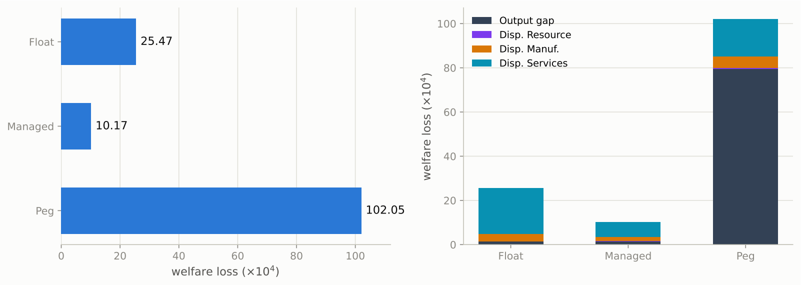
Export/ToT shock welfare ($\times 10^4$), old$\rightarrow$new: Float $0.11 \rightarrow 0.17$, Managed $0.18 \rightarrow 0.14$, Peg $2.16 \rightarrow 1.17$

Ranking preserved: $10.96 < 25.15 < 90.45$ once exports are concentrated in the true exporting (Resource) sector

Concentrating exports in the flexible-price sector lowers Peg's ToT cost: Resource ($\delta_1 = 0.90$, low κ_1) vs. Services (stickiest, high κ_3) cuts Peg's ToT welfare cost 46%

Caveat: matching export *shares* exactly shrinks total trade volume $\approx 22\%$ ($0.256 \rightarrow 0.201$); $BSTAR_{ss} = 0$ still holds exactly by construction

Welfare Loss: Total & Decomposition



Managed float dominates both corners. Peg's loss is ≈78% output gap (79.53 of 102.05, zero monetary autonomy); Float's and Managed's are almost entirely dispersion.

Mechanisms: Why Each Regime Costs What It Costs

Regime	Cost channel	What sets the magnitude
Float	S_t's near-unit-root persistence (NFA state) feeds $\Gamma_i \Delta e_t$ into *every* sector's MC, even for shocks with no FX content — pure dispersion cost	Persistence of debt-elastic premium (ψ); Γ_i (import exposure); κ_i of the most central/sticky sector (Services: 20.7 of 25.5)
Peg	Zero monetary autonomy: output-gap term explodes (79.5 of 102.1, $\approx$78%) on the UIP shock; $\Delta e_t \equiv 0$ *relocates*, not removes, adjustment into upstream P^H	Absence of ϕ_π, ϕ_y; real vs. nominal share of the shock; upstream sector's Domar weight
Managed	$\phi_S \Delta \log S_t$ damps $\Gamma_i \Delta e_t$ without fully re-imposing peg's relocation cost — dominates every sector at once	Interior optimal $\phi_S^* \approx 0.20$: too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ autonomy lost

Two forces, always in tension: Γ-channel favors peg (switches off $\Gamma_i \Delta e_t$); **relocation** channel favors float (peg's fixed e_t forces the same adjustment into upstream prices). Which wins depends on Γ_i vs. κ_i; managed is the only regime improving on *both* corners for every sector.

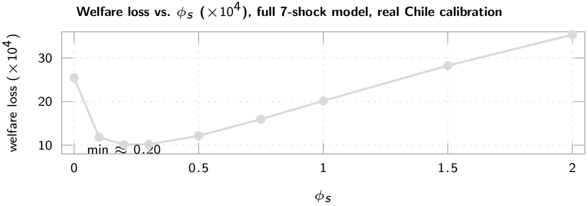
Analytical or simulated? Both.

Channels — closed-form: $\Gamma_i = [\tilde{A}\boldsymbol{\omega}_F]_i/\kappa$ (pass-through), $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ (welfare weight), DC-survival $\boldsymbol{\phi}^\top \mathcal{V}=\mathbf{0}^\top$ *and* $\boldsymbol{\phi}^\top \Gamma=0$ — from primitives alone, no model solve.

Magnitudes/ranking — need simulation: $\mathrm{Var}(\pi_{it}), \mathrm{Var}(\tilde{y}_t)$ come from the full linear RE solution (order-1 Dynare) — no closed form once $\phi_S>0$ or Ω^H dense.

Managed$<$Float$\ll$Peg and $\phi_S^*\approx$0.20 are simulated, but robust across all four calibrations.

Managed Float: Optimal Stabilization Intensity ϕ_s



Why interior: too low → UIP shock leaks in (ϕ_s=0 collapses to Float's 25.47); too high → loses autonomy

ϕ_s=0 → 0.20: loss 25.47 → 10.14. **Interior optimum, but nearly flat near it:** 0.20 beats the calibrated 0.30 by only ≈0.3% (was ≈13% under the old stylized-network sweep — the Chile calibration's optimum is much less sharply peaked)

Beyond ≈ 0.3: rises steadily, back above Float (25.47) between ϕ_s=1.0 (20.16) and 1.5 (28.26)

Appendix: DC Index — Definition and Insulation

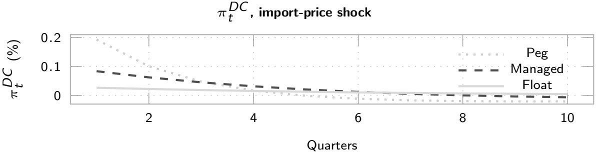
Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$ rank-deficient $\Rightarrow$ unique DC index:

$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it},\ w_i^{DC} \propto \lambda_{D,i} \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Open economy: 2 channels (TFP + FX):

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \textbf{generically infeasible}$$

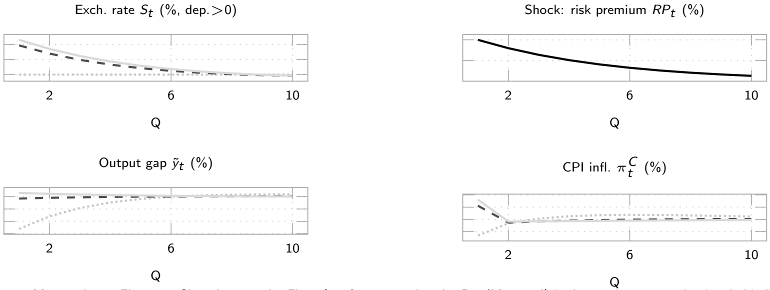
Knife-edge test: $\Gamma \propto \mathcal{B}$ would save the DC index. Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i/\mathcal{B}_i$ shrinks $\approx$2$\times$ per sector, not constant. Breakdown is real, not a numerical artifact.
Fix: import nodes flexible ($\hat{\delta}_M$=1 $\Rightarrow$ w_M=0) $\Rightarrow$ DC well-defined again by construction — what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown below.



Float: smallest on impact (0.026%) & most persistent; Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

IRFs: Foreign Risk-Premium Shock — the Key Mechanism

The shock that decides the ranking: 75% of Peg's welfare loss (Chile calibration).

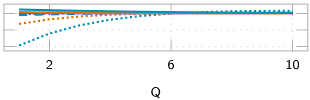


Peg ••• Managed - - Float —. Slow decay under Float (vs. fast reversal under Peg/Managed) is the persistence mechanism behind Float's larger sectoral dispersion cost. Sector-level IRFs: Appendix (next slide).

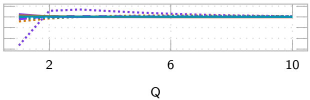
IRFs: Foreign Risk-Premium Shock — Sectoral Detail

Sector-level counterpart to the previous risk-premium IRF slide — underlies the sectoral dispersion costs on the Network Exposure & Welfare slides.

Sectoral output gap, all regimes (%)

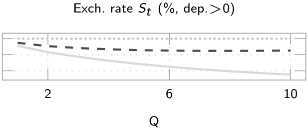


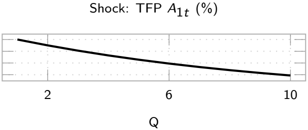
Sectoral inflation, all regimes (%)

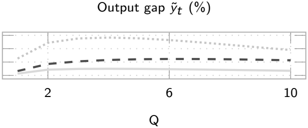


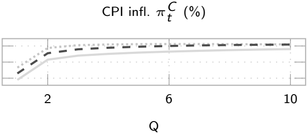
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

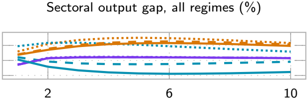
IRFs: Resource TFP Shock

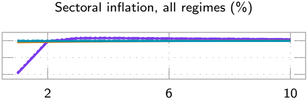






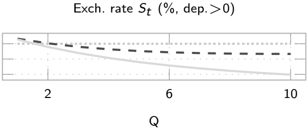


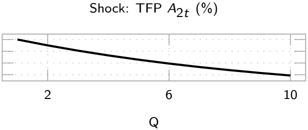


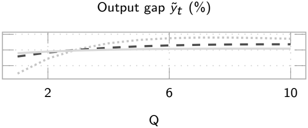


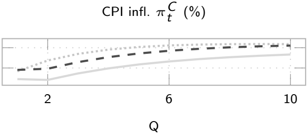
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

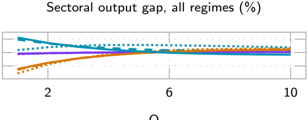
IRFs: Manufacturing TFP Shock

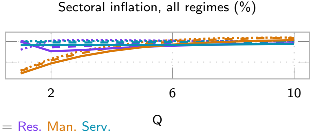








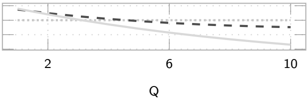




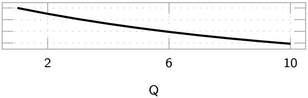
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Services TFP Shock

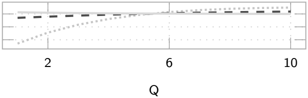
Exch. rate S_t (%, dep.>0)



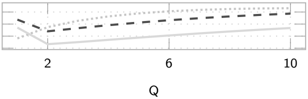
Shock: TFP A_{3t} (%)



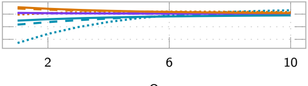
Output gap $\tilde{y}_t$ (%)



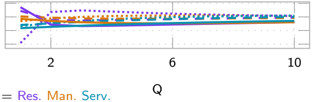
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)

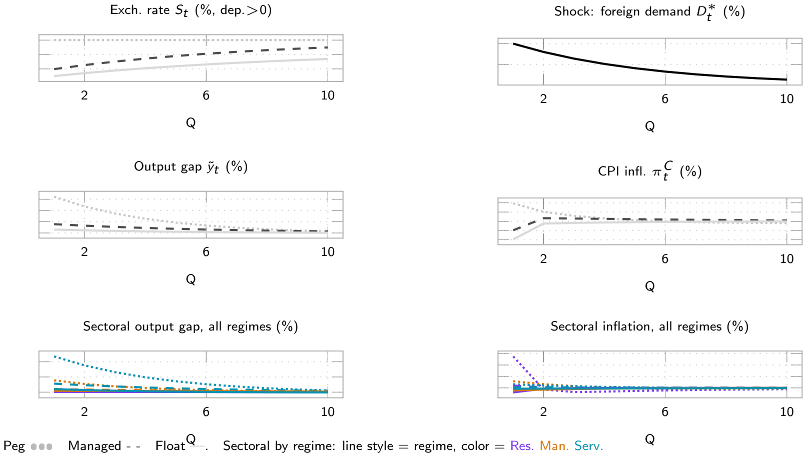


Sectoral inflation, all regimes (%)



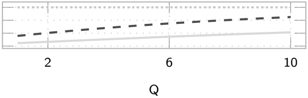
Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Foreign-Demand Shock

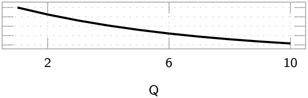


IRFs: Export-Price / ToT Shock

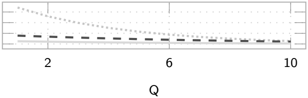
Exch. rate S_t (%, dep.$>$0)



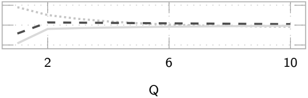
Shock: export price P_t^X (%)



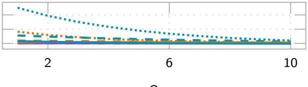
Output gap $\tilde{y}_t$ (%)



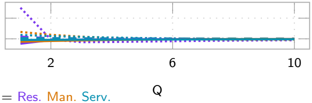
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)

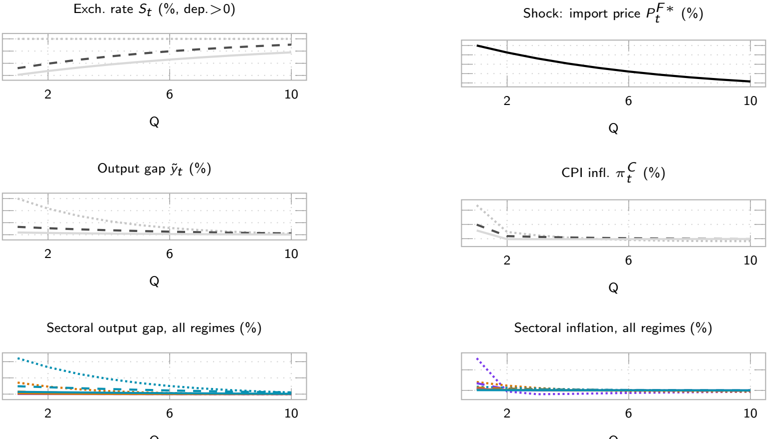


Sectoral inflation, all regimes (%)



Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Import-Price Shock (Backup)



Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

Appendix: Model Equations & Notation (1/2)

Households

$$\max\ \mathbb{E}_0 \sum_t \beta^t \Big[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \Big] \quad \text{s.t.} \quad P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)] S_t B_{t-1}^*$$

$$C_t^{-\gamma} = \beta(1+i_t)\mathbb{E}_t[C_{t+1}^{-\gamma}/\Pi_{t+1}^C], \quad W_t/P_t^C = C_t^{\gamma} L_t^{\varphi}, \quad (1+i_t) = (1+i^*)[1-\psi(\hat{b}_t^* - \bar{b}^*)]\ \mathbb{E}_t[S_{t+1}/S_t]\ \text{(UIP)}$$

C, L: consumption, labor — β, γ, φ: discount, RRA, inv. Frisch — W, i: wage, dom. rate — S, i^*: ER, foreign rate
ψ: debt-elastic premium — B^*: foreign bond — P^C: CPI — Π_t: profits (rebated)

Firms: Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \quad \Rightarrow \quad MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A: output, TFP — L, X, M: labor, dom./imp. inputs — Ω_{ij}^H: dom. IO share — Ω_i^F: import cost share
α_i: labor share — MC_{it}: nominal marg. cost — P^{F*}: foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Appendix: Model Equations & Notation (2/2)

Firms: Calvo Pricing — reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta\Big(\frac{C_{t+1}}{C_t}\Big)^{-\gamma}\Pi^{\varepsilon}_{i,t+1} X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1-\delta_i)\beta\Big(\frac{C_{t+1}}{C_t}\Big)^{-\gamma}\Pi^{\varepsilon-1}_{i,t+1} X2_{i,t+1}$$

$$P^{\#}_{it}/P_{it} = \tfrac{\varepsilon}{\varepsilon-1}\, X1_{it}/X2_{it}, \quad 1 = (1-\delta_i)\Pi^{\varepsilon-1}_{it} + \delta_i (P^{\#}_{it}/P_{it})^{1-\varepsilon}, \quad \hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$$

δ_i: Calvo reset prob. (primitive) $\hat{\delta}_i$: derived slope (enters DC weights) $P^{\#}_{it}$: reset price ε: elasticity (markup $\varepsilon/(\varepsilon-1)$)

Open Economy

$$\underbrace{P^{FH}_t = S_t P^{F*}_t}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*)\left[1-\psi(\hat{b}^*_t - \bar{b}^*)\right]\mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}^*_t = \frac{1+i_{t-1}}{\Pi^C_t}\hat{b}^*_{t-1} + \frac{P^H_t EX_t - P^{FH}_t IM_t}{P^C_t}, \quad EX_t = \kappa_{EX} D^*_t \Big(\frac{P^H_t}{P^{FH}_t}\Big)^{-\theta^*}$$

$$Y_{it} = \beta^H_i \frac{P^H_t}{P_{it}}(C^H_t + EX_t) + \sum_j \Omega^H_{ji}\frac{MC_{jt}Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled) EX, IM: exports, imports θ^*: export elasticity D^*: foreign demand shifter
β^H_i: cons. share, sector i Float: i_t Taylor rule, S_t free Peg: $S_t = \bar{S}$, i_t residual

Appendix: The Peg vs. Float Debate — What Do We Know?

Why it depends (structure):

- **Mundell (1961)** OCA — fix favored by small, open, factor-mobile economies w/ correlated shocks vs. anchor; float favored by large, diversified ones
- **Frankel (1999)** — "no single regime is right for all countries or at all times"; openness, size, trade & capital-market structure all matter

Shock-absorption view (why float wins on real shocks — robustly):

- **Friedman (1953)** — S_t adjusts instead of costly wage/price deflation
- **Broda (2004)** — 75 dev. countries, 1973–96: negative ToT shocks $\rightarrow$ large real GDP declines under pegs
- **Edwards & Levy-Yeyati (2005)** — 10% ToT deterioration: -0.80pp growth under pegs vs. -0.43pp under floats
- **Céspedes, Chang & Velasco (2004)** — even with dollarized-liability balance-sheet effects, float *still* beats fixed

But floating is not a free lunch (limits on float's insulation):

- **Calvo & Reinhart (2002)** "Fear of Floating" — EMEs peg de facto anyway: credibility-building, not literal welfare optimum
- **Rey (2013)** "Dilemma not trilemma" — under free capital mobility, float alone doesn't deliver monetary autonomy

Consensus: no corner solution is universally optimal — the choice trades a nominal anchor against real-shock absorption, shock- and structure-dependent.

Appendix: The Frontier — Is Fear of Floating Actually Optimal?

Itskhoki & Mukhin (2023) — general open-economy framework with endogenous UIP/PPP deviations: efficient policy = MP closes the output gap + FX intervention eliminates the UIP wedge. Optimal policy is generically a **managed float/crawling peg**, matching observed fear of floating. When the natural RER is stable, this collapses to an "open-economy divine coincidence."
↪ *vs. us:* **same headline** (Managed, not a corner, is optimal) — reached instead via a multi-sector Calvo network with a persistence-driven, not UIP-gap-closing, mechanism

Bianchi & Coulibaly (2025; 2022 WP) — fear of floating is **optimal**, not behavioral, when credit access is tied to the real exchange rate (self-fulfilling crisis risk). Prescription: float for real shocks, **fix** for financial shocks
↪ *vs. us:* **opposite ranking** for our financial shock — our risk-premium/UIP shock passes straight through under **Peg** (worst regime). Different financial friction (persistent NFA wedge vs. collateral-constraint crisis) flips the prescription

Coulibaly (2023) — discretionary MP is procyclical to offset balance-sheet effects; commitment (inflation targeting) or capital controls raise welfare
↪ *vs. us:* echoes our "Managed = extra instrument" logic — giving policy more tools beats a corner regime

Bottom line: the 2023–25 frontier has also moved past corner solutions — but *why* fear of floating is optimal is mechanism-specific. We add production-network amplification + shock persistence as a distinct reason, not examined by either paper.

Appendix: How Our Results Compare to the Literature

Literature	Says	Us
Mundell (1961); Frankel (1999)	Optimal regime depends on structure, no universal answer	**Confirms**: ranking flips with shock type (UIP vs. TFP) and network density
Friedman (1953); Broda (2004); Edwards–Levy-Yeyati (2005); Céspedes–Chang–Velasco (2004)	Float buffers real (ToT/foreign) shocks, lower output loss than peg — robust even to balance-sheet effects	**Same direction** (Peg worst), **different shock**: risk-premium/UIP, not ToT
Galí & Monacelli (2005)	Peg dominated, float ≈ optimal in 1-sector SOE	**Same ranking of Peg**, but pure **Float is not optimal**: Managed beats it
Calvo–Reinhart (2002)	EMEs avoid pure float — read as **fear**, a credibility-driven deviation from optimum	We find partial stabilization is **literally welfare-optimal** — a persistence-driven, not fear-driven, reason
Rey (2013)	Float alone doesn't insulate; need a second instrument (capital controls/macropru)	Echoes Fanelli–Straub: Managed uses S_t itself as that **second instrument**
Qiu, Wang, Xu & Zanetti (2026, JME)	Open-economy DC index (CPI+expenditure-switching+profit channels), *static*, WIOD-calibrated; DC breaks down, output-gap targeting near-optimal	**Independent corroboration** DC breaks down in open economy; **we add** the piece they flag as their own top open extension — UIP, NFA persistence, and FX-**regime choice**

Novel angle: the classic shock-absorber logic (Friedman/Broda) is about shock **type**

Appendix: The ECB's Open-Economy Production-Network Model

Gnocato, Montes-Galdón & Stamato (2025), "Tariffs across the supply chain," ECB WP No. 3081 — also underlies Lane's (ECB Exec. Board) 13 May 2026 speech on energy shocks (4-region extension):

- **Setup:** two-region, multi-sector, full IO structure, international production network; sticky wages/prices; Taylor rule on *domestic* CPI (no regime comparison)
- **Finding:** tariffs on **intermediate** inputs → persistent inflation & larger GDP loss (low substitutability, network propagation); tariffs on **final** goods → one-off inflation spike, smaller GDP loss — *where* in the network a shock hits sets its persistence
- ↪ *vs. us:* their tariff-location result and our TFP-vs-UIP result are the same insight in two settings: shock **location/type** sets its persistence, echoing our "driver is persistence, not volatility"

What it cites for open economy + production networks:

- Network amplification, closed economy: **Pasten, Schoenle & Weber (2020)**; **Ghassibe (2021)**; **Rubbo (2023)**[2]; **Afrouzi & Bhattarai (2023)**
- Open-economy multi-sector NK + networks: **Devereux, Gente & Yu (2023)**; **Hinterlang et al. (2023)** EMuSe (Bundesbank); closest: **Kalemli-Özcan, Soylu & Yildirim (2025)** "Global networks, MP and trade"
- Trade fragmentation & inflation: **Moro & Nispi Landi (2024)**; **Ambrosino, Chan & Tenreyro (2026, JIE)**
- Optimal MP under tariffs: **Bergin & Corsetti (2023)**; **Bianchi & Coulibaly (2025)** "Optimal MP response to tariffs" (*different* paper from their fear-of-floating one)

Bottom line: none compares FX **regimes** or asks whether a DC index survives — they fix the monetary rule. That gap is still exactly our question.

[2] Published version: Rubbo, E. (2023), Econometrica 91(4), 1417–1455 — journal year is 2023, not 2024.